

Smart Traffic Lights using Local AI

Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually follow fixed timers. They are simple and inexpensive, but they cannot easily react when one road suddenly becomes crowded. Recent systems use cameras and artificial intelligence (AI) to estimate traffic conditions and adjust traffic-signal timing. Research reviews show growing interest in AI-based traffic signal control, including deep reinforcement learning approaches for adapting signal control to complex traffic conditions [1].

Cloud-based AI systems send camera information to a central server for processing. This can provide strong computing power, but it depends on an internet connection and can introduce communication delay. Local smart-camera systems move some processing closer to the road. This can reduce the amount of information that must be sent over the internet and may allow faster decisions. A small local computer can examine camera input and send a short instruction to the traffic-light controller. Machine-vision research has also explored using camera-based vehicle information to adapt traffic-light timing according to traffic flow and vehicle conditions [2].

The systems can be compared using three simple measures. Basic Timed Lights make decisions every 30 seconds, use zero internet data, and have a setup cost of \$200. Cloud AI Systems make decisions in 2 seconds, use high internet data, and have a setup cost of \$2,500. Local Smart Cameras make decisions in 0.5 seconds, use low internet data, and have a setup cost of \$1,200.

Simple System Architecture

The proposed architecture follows four steps. First, Street Cameras capture traffic conditions. Second, the video information is sent to Smart Pole Computers, which perform local processing and make a quick traffic decision. Third, selected information can be forwarded to the City Cloud Server for storage and wider analysis. Fourth, the decision is sent to the Traffic Lights, which change their signals. The flow is therefore: **Street Cameras** → **Smart Pole Computers** → **City Cloud Server** → **Traffic Lights**.

Research Gap

Existing traffic systems can work well for regular daily traffic, especially when traffic patterns are predictable. However, an important emergency problem remains: traffic signals must also be able to respond quickly when an emergency vehicle such as an ambulance approaches. Research on emergency-vehicle signal control has investigated real-time signal strategies and signal coordination to provide faster passage through intersections [3], [4]. This creates a need for a smart traffic system that can detect an emergency vehicle and quickly coordinate nearby signals to create a clear route.

References

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